Project Planning

Project Integration Management

- Project Integration involves unification, consolidation, articulation, and actions that are required for project completion, meeting customer requirements, and managing expectations
- Involves making choices about where to concentrate resources and effort on any specific day
- Anticipating potential issues, dealing with them before they become critical, and coordinating the work for the overall good
- Making trade-off among competing objectives and alternatives

Need for Integration

- As the processes are interacting, there is a need for integration management
- For example, a cost estimate needs the integration of planning processes, Project cost management processes, project time management processes and project risk management processes
- The project deliverables also need to be integrated with ongoing operations of either the performing organization or customer’s organization

Major Project Management Integration Processes

- Develop a project charter that formally authorizes a project
- Develop Preliminary Project Scope Statement
- Develop Project Management Plan
- Direct and Manage project execution
- Monitor and control project work
- Integrated change control
- Close the project

Develop Project Charter

- Project charter is a document that formally authorizes a project
- The project sponsor external to project organization issues the project charter
- Projects are chartered and authorized by an enterprise, a Government agency, or a company as a result of one or more of the following reasons

Methods for Selecting Projects contd..

- A market demand
- A business need
- A customer request
- A technological advance
- A legal requirement
- A social need

All of the above are also called problems, opportunities or business requirements

Project charter contd..

- Chartering a project links the project to the ongoing work of the organization
- Developing a project charter is concerned with documenting the business needs, project justification, current understanding of the customer requirements, and the new product or service or result that is intended to satisfy those requirements

Project charter contd..

The project charter should address the following information

- Requirements that satisfy customer, sponsor, and other stakeholder needs, wants and expectation
- Business needs, high-level project description, or product requirements that the project is undertaken to address
- Project purpose or justification
- Assigned project manager and authority level
- Summary milestone schedule

Project charter contd..

- Stakeholder influences
- Functional organizations and their participation
- Organizational, environmental and external assumptions
- Organizational, environmental and external constraints
- Business case justifying the project, including return on investment
- Summary budget

Project Plan Development

- Project management plan includes actions necessary to define, integrate, and coordinate all subsidiary plans that define how the project is executed, monitored , and closed
- These subsidiary plans include
- Project scope management plan
- Schedule management plan
- Cost management plan
- Quality management plan
- Process improvement plan

Project Plan Development contd..

- Communication management plan
- Risk management plan
- Procurement management plan
- Milestone list
- Resource Calendars (identifies resource specific holidays and resource availability periods)
- Schedule Baseline (it has schedule start dates and schedule finish dates)
- Cost Baseline (basis of comparison)
- Quality Baseline
- Risk Register

Project plan execution

- The direct and manage project execution process requires the project manager and his team to perform actions to execute the project management plan to accomplish that work defined in the scope statement
- Some of these actions include
- Perform activities to accomplish project objectives
- Train and manage the project team members assigned to the project
- Obtain quotations, bids, offers, or proposals as appropriate

Project plan execution contd..

- Select sellers from choosing from among potential sellers
- Obtain, manage and use resources including materials, tools, equipment, and facilities
- Implement the planned methods and standards
- Create, control, verify and validate project deliverables
- Manage risks and implement risk response activities
- Manage sellers
- Adapt approved changes into the project’s scope, plans and environment

Project plan execution contd..

- Establish and manage project communication channels both external and internal to the project team
- Collect project data and report cost, schedule, technical and quality progress and status information to facilitate forecasting
- Collect and document lessons learned and implement approved process improvement activities

Project plan execution contd..

- Information about the completion status of the deliverables and what has been accomplished is collected as a part of project execution and is fed into the performance reporting process

- Though the products, services or results of the project are frequently in the form of tangible benefits as buildings, roads etc, intangible deliverables as training can also be provided

Project plan execution contd..

- Project plan execution also requires implementation of the following
- Approved corrective actions that will bring anticipatory project performance into compliance with project management plan
- Approved preventive actions to reduce the probability of potential negative consequences
- Approved defect repair requests to correct product defects found by the quality process

Project scope management

- Project scope management (PSM) is concerned with defining and controlling what is and what is not included in the project
- PSM includes the processes required to ensure that the project includes all the work required to complete the project successfully

Project scope management contd..

- PSM includes the following
- **Scope planning**: Creating a plan that documents how the project scope will be defined, verified, controlled and how the work breakdown structure (WBS) will be created and defined
- **Scope definition:** Developing a scope statement as the basis for future project decisions
- **Create work breakdown structure**: Subdividing the major project deliverables and project work into smaller, more manageable components
- **Scope verification**: Formalizing acceptance of the completed project deliverables

Project scope management contd..

- **Scope control**: Controlling changes to the project scope
- **Scope planning:** Defining and managing the project scope influences project's overall success

  The project scope management plan is a document that describes project scope, detailed project scope statement, define and develop WBS, verify and control project scope
- **Detailed Scope definition:** The detailed scope statement depends on major deliverables, assumptions, constraints, stakeholder needs and expectations

  It is a refinement of preliminary project scope statement as more information about the project is known

  The assumptions and constraints are analysed and more assumptions and constraints are added as necessary

Project Scope Statement

- Project scope statement (PSS) describes the project's deliverables and objectives and also the work required to create those deliverables
- The degree and level of detail PSS defines the work to be performed and the work to be excluded can determine how well the project management team can control the overall project scope
- The detailed PSS includes directly or indirectly the following documents

Project Objectives: It Includes business, cost, schedule, technical and quality objectives

Project Scope Statement contd..

Product scope description: Describes the characteristics of the product, service or result that the product was undertaken to create

Project Requirements: Describes the conditions or capabilities that must be met by the deliverables

Stakeholder needs, and expectations are translated into prioritized requirements

Project Boundaries: It states what is included in the project and explicitly states what is excluded from the project; otherwise the stakeholder might assume that a particular product, service or result could be a component of the project

Project Scope Statement contd..

- **Project deliverables:** Depending on the project scope statement, the deliverables are described at a summary level or at a great detail
- **Product acceptance criteria**: Defines the process and criteria for accepting completed products
- **Process constraints:** More detailed than the constraints included in the project charter; schedule milestones, predefined budget or contractual provisions etc.
- Project assumptions: The assumptions are more detailed than in the project charter

Project assumptions contd..

- The assumptions include

Project organization

Defined risk

Schedule milestones

Fund limitation

Cost estimate

Project configuration management requirements

Change control

Project specifications

Project approval requirements

Work Breakdown Structure

- WBS is a deliverable oriented hierarchical decomposition of the work to be executed by the project team to accomplish the project objectives and create the required deliverables
- The WBS subdivides the project work into smaller, more manageable and detailed pieces of work
- Decomposition of the total project work generally involves the following activities

- Identifying the deliverables and related work
- Structuring and organizing the WBS

Work Breakdown Structure contd..

- Decomposing the upper WBS levels into lower level detailed components
- Developing and assigning identification codes to the WBS components
- Verifying that the degree of decomposition of the work is necessary and sufficient
- The resulting WBS can take a number of forms

  - Using the major deliverables and subprojects as the first level of decomposition

  - Using subprojects as the first level of decomposition, the subprojects may be contracted out; the contractor applies WBS to contracted work

Work Breakdown Structure contd..

- Using the phases of the project life cycle as the first level of decomposition, the project deliverables inserted at the second level
- Example WBS for defence aircraft system

Aircraft system

Training	Air vehicle	Data	Test and Evaluation
Equipment TR	Airframe	Technical data	Mockups
Facility TR	Engine	Engineering data	Operational Test
Service TR	Communication system	MgMt data	Development
	Navigation system		Test
	Fire control system		